המחלקה לטכנולוגיות דיגיטליות ברפואה
Department of Digital Medical Technologies

מכון טכנולוגי חולון
Holon Institute of Technology

More than 50 YEARS of Academic Excellence

Analysis and Classification of Histological Images of Colorectal Cancer

Final Project - Machine Learning

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Lucturers - Mr. Yakir Menachem and Mr. Ron Ben David

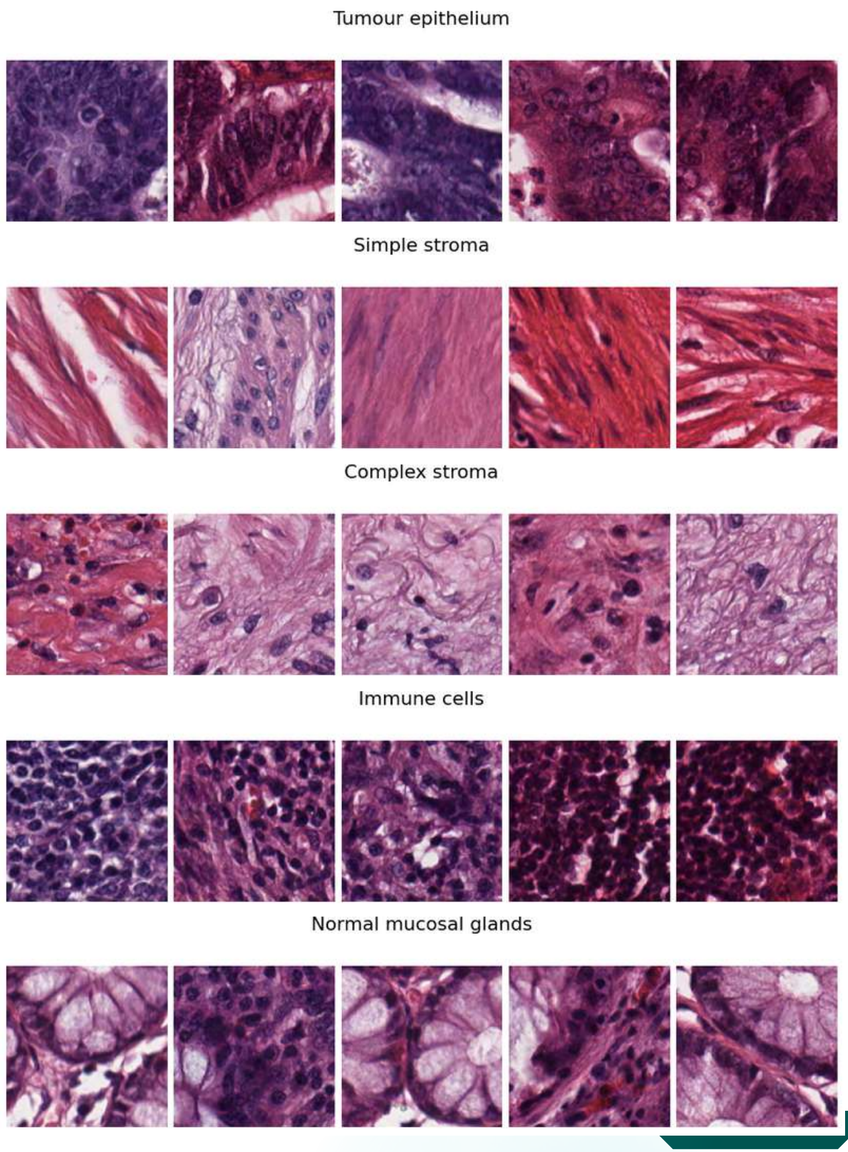
Summer Semester 2025 תשפ״ה

Digital Medical Technologies · Digital Medical Technologies

Dataset and Project Goal

Dataset

- Public Kather Texture 2016 dataset
- 5,000 RGB histological image patches of colorectal tissue
- Each image size: 150×150 pixels
- Each patch belongs to one of 8 tissue classes: Tumor epithelium, Simple stroma, Complex stroma, Immune cells, Debris, Normal mucosal glands, Adipose tissue, Background

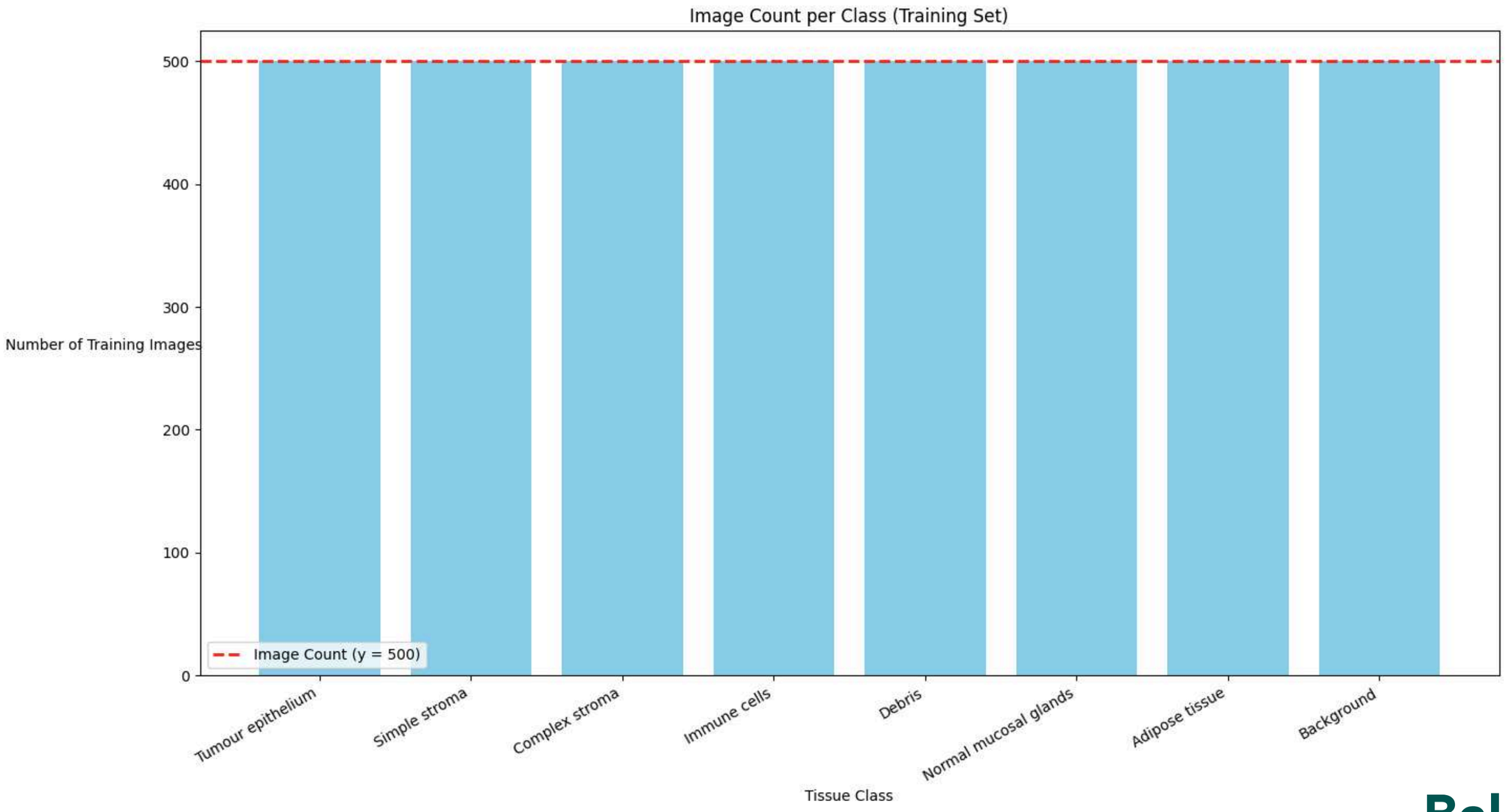


Goal

develop and evaluate an automated pipeline for classifying colorectal cancer histology images into eight predefined tissue classes by comparing alternative feature-extraction methods and modeling strategies.

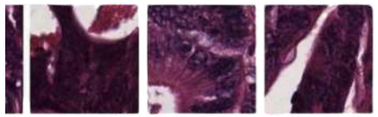
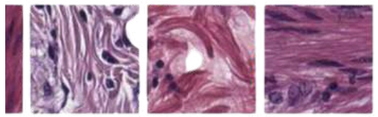
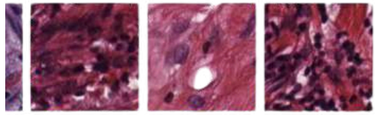
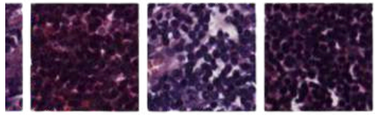
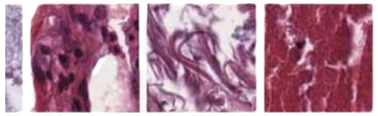
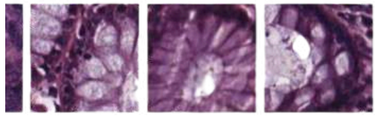
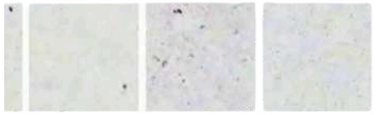
Exploratory Data Analysis

EDA – Label Distribution

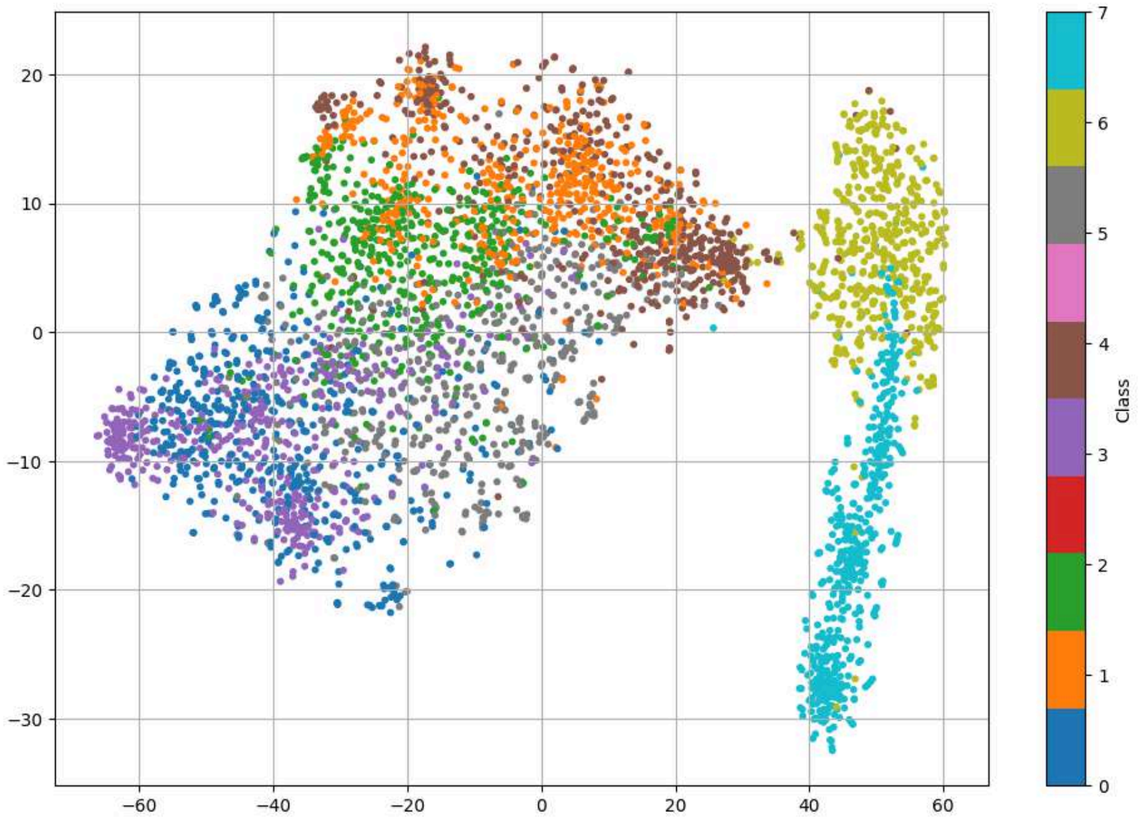


Balanced!

EDA – Sample Images

אפיתל של גידול		Tumour Epithelium
רקמת חיבור בסיסית		Simple Stroma
תאי גידול / תאי חיסון		Complex Stroma
תאי T/B ולימפוציטים		Immune Cells
פסולת - רקמה נמקית, דימומים וכו׳		Debris
בלוטות תקינות של רירית		Normal Mucosal Glands
רקמת שומן		Adipose Tissue
ללא רקמה פעילה		Background (no tissue)

PCA → t-SNE Visualization

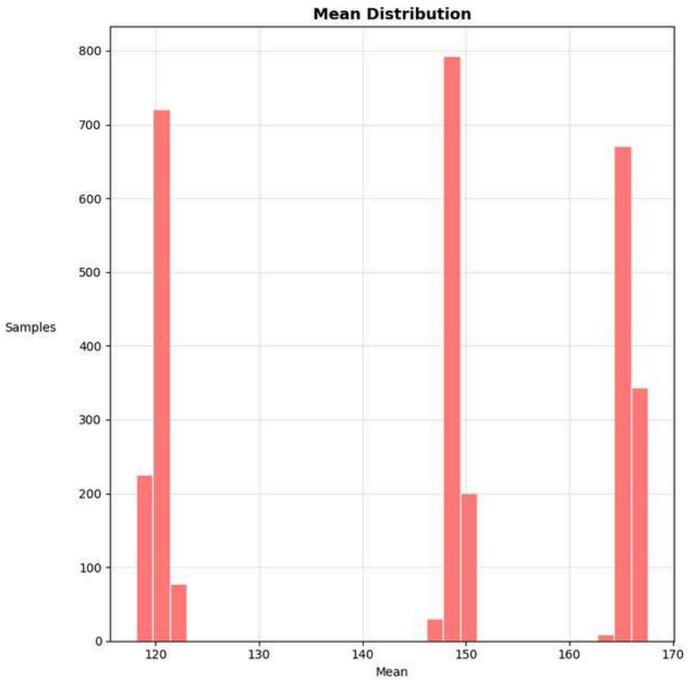


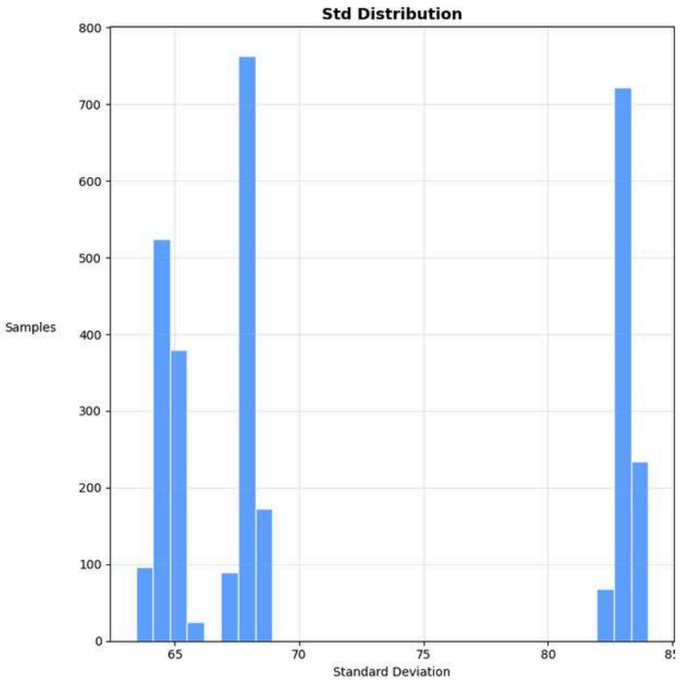
Partial class separation with notable overlap emphasizing the challenge and the need for strong feature extraction

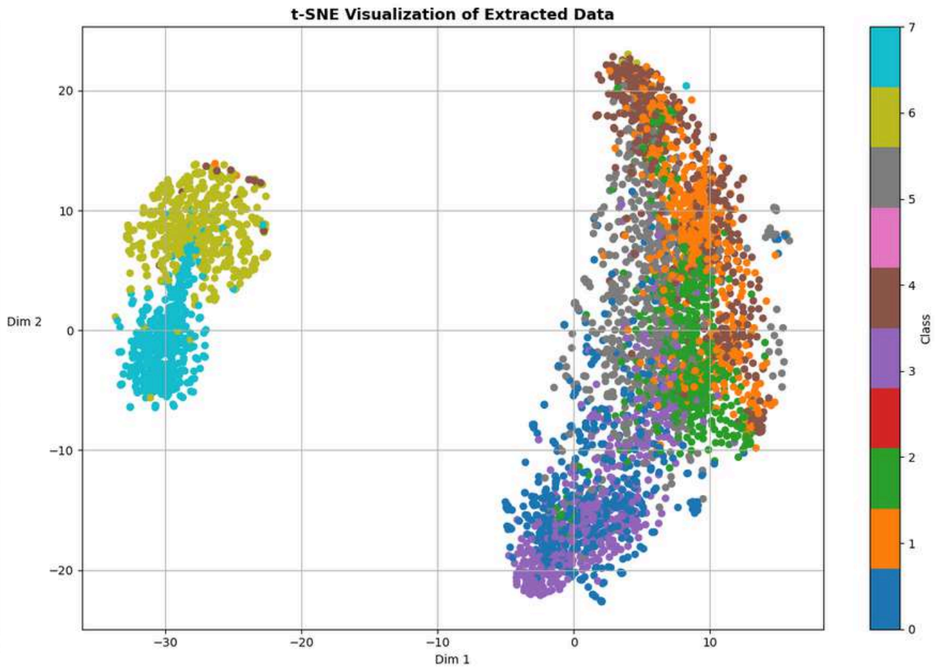
150x150x3 → PCA → 50D → t-SNE → 2D

Downsample & Flatten Features

Method - Each image was downsampled to 32x32 and flattened into a 1x3072 feature vector







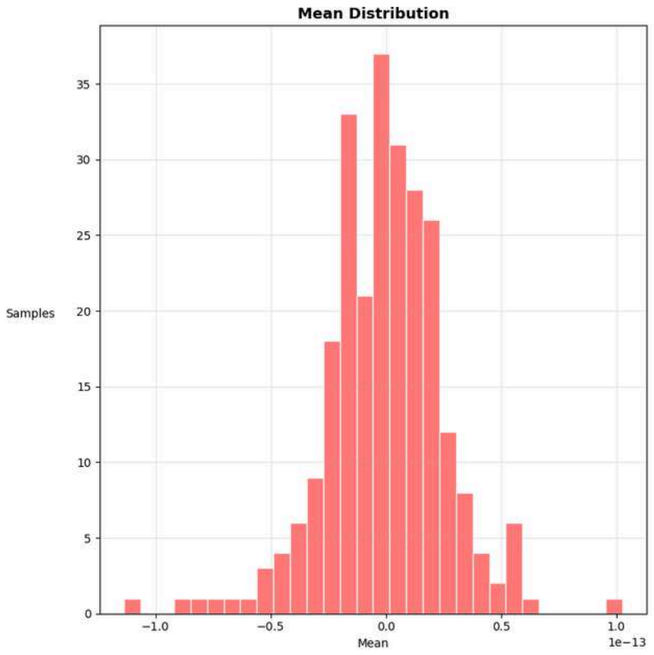
Feature means: 145.0

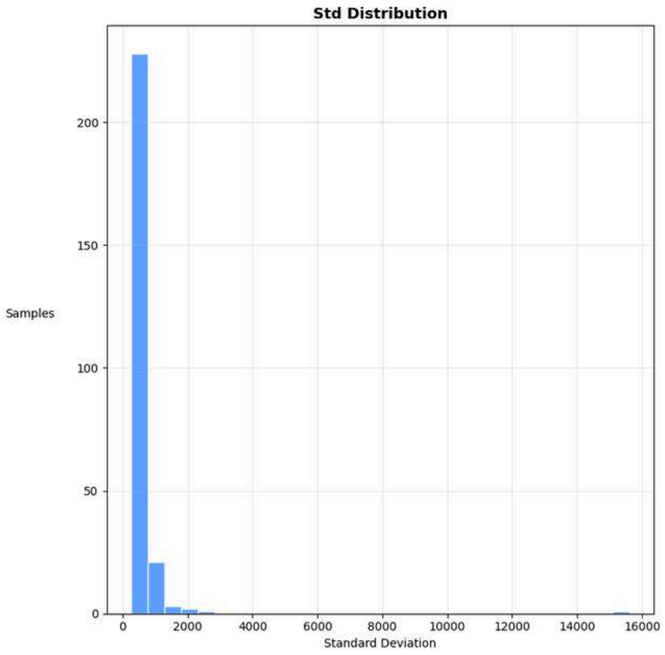
Standard deviations: 71.912

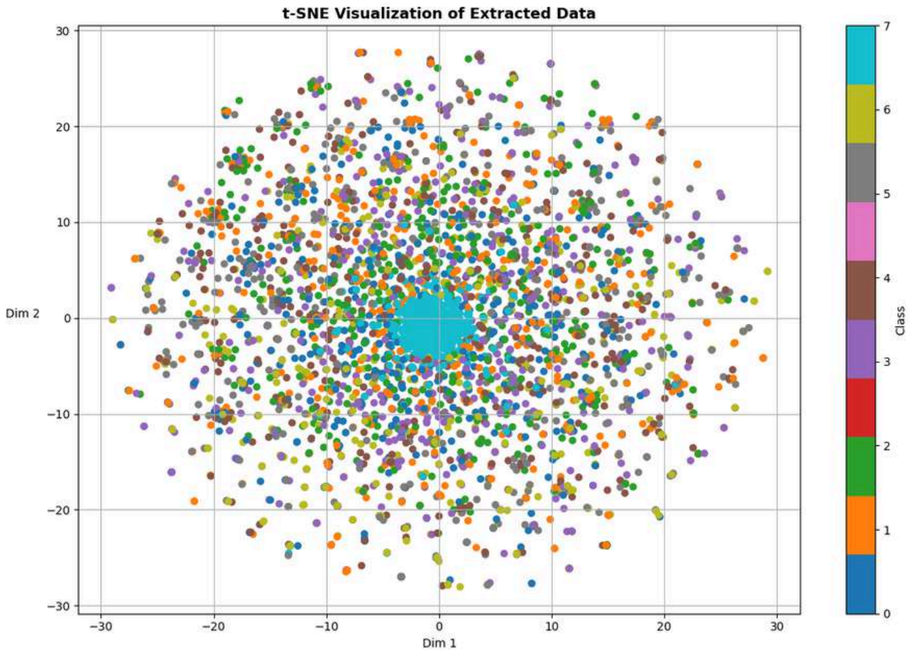
Normalization is required

PCA Features

Method - PCA was used to reduce the flattened 150x150 images to 256 prior principal components







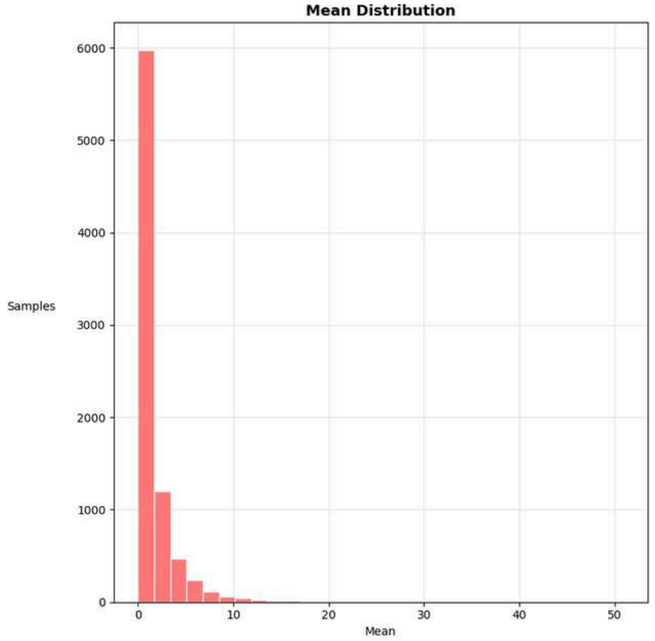
Feature means: 0.0

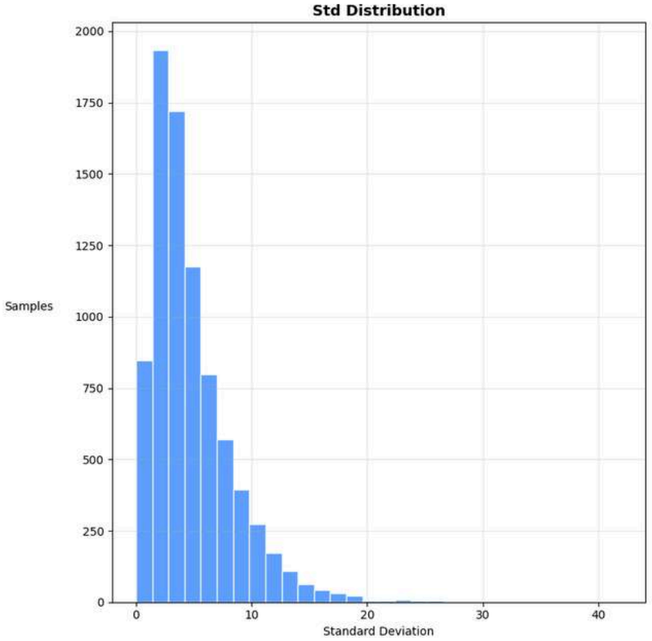
Standard deviations: 555.124

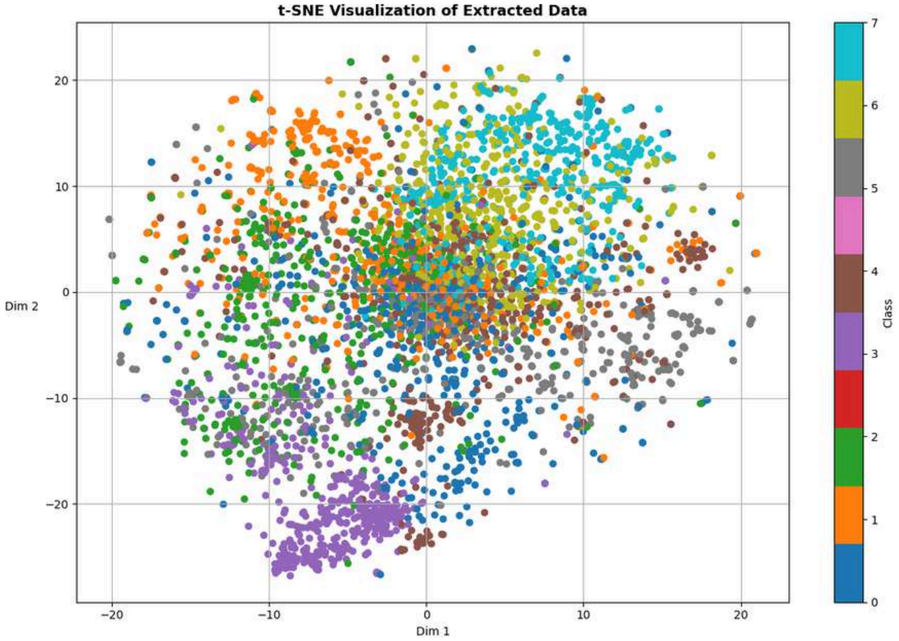
Normalization is required

VGG16 Backbone Features

Method - A pre-trained VGG16 model was used to convert each image into a high-dimension feature vector (1x8192)







Feature means: 1.598

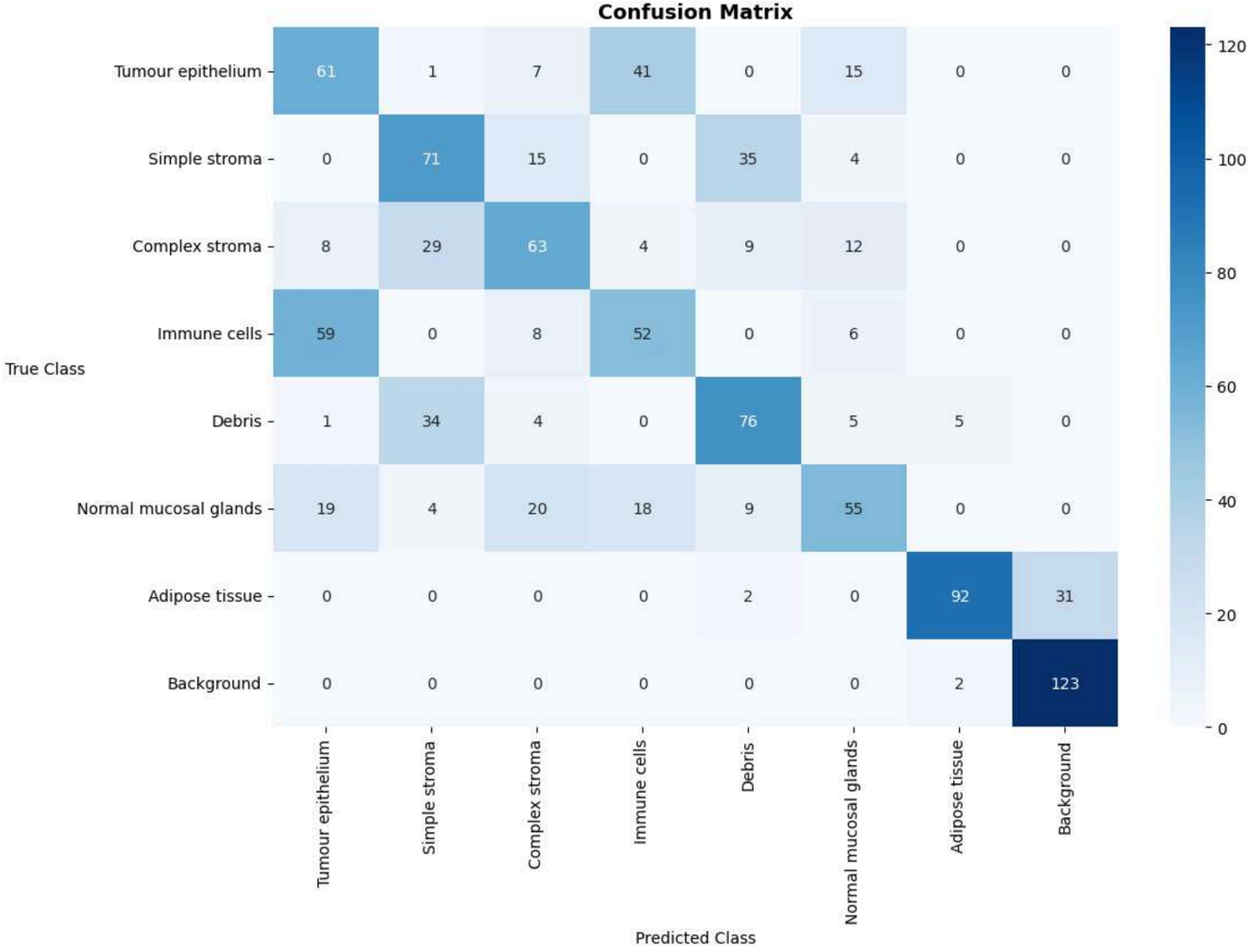
Standard deviations: 4.816

Normalization is required

Model Performance

1 Downsample + SVM (linear)

Model: Achieves high training accuracy with basic texture features



Evaluation Report:

Training Accuracy: 0.9995

Testing Accuracy: 0.593

Training F1 Score: 0.9995

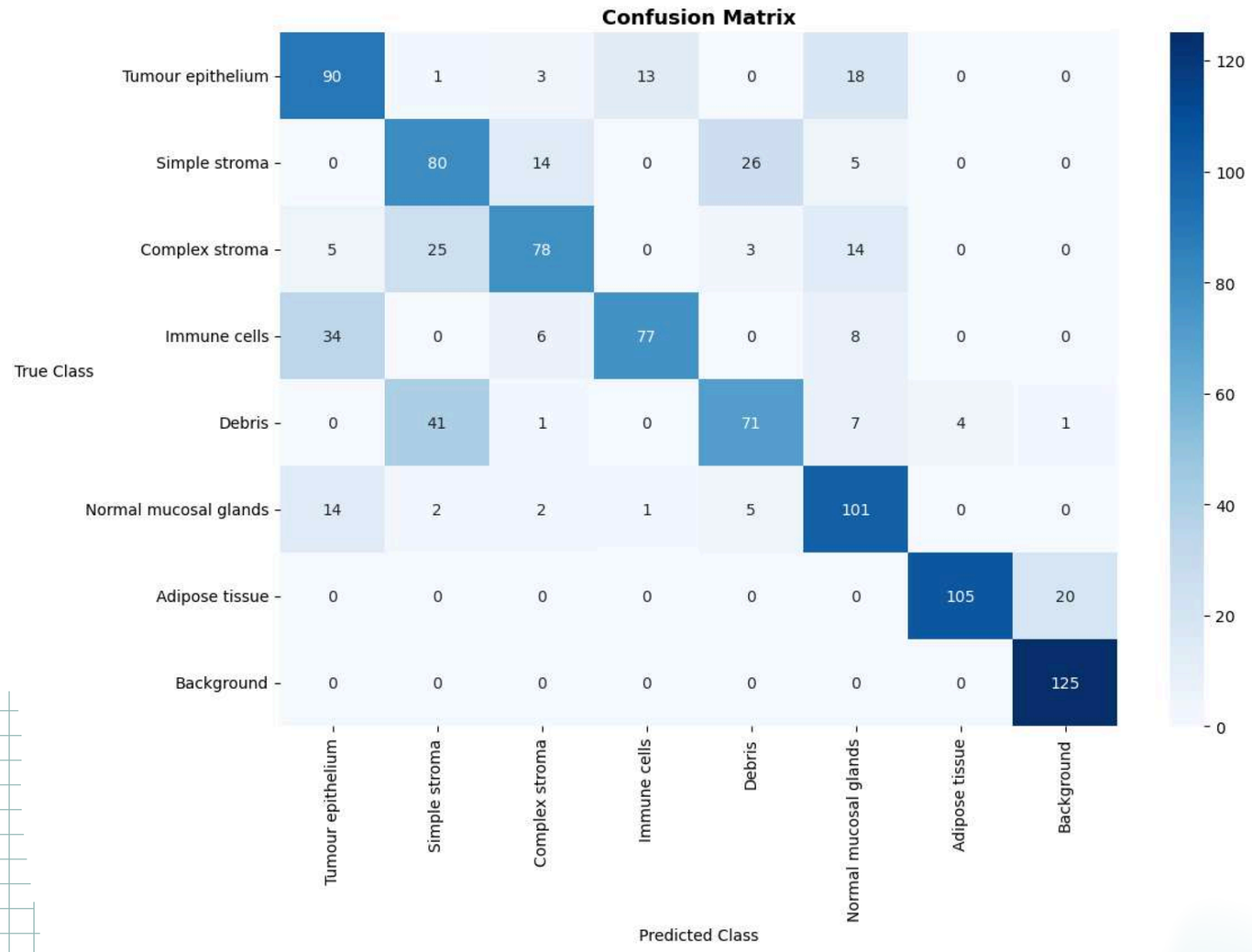
Testing F1 Score: 0.5914

Overfitting Check: 0.407

Clear overfitting, poor generalization on test data

1 Downsample + SVM (RBF)

Model: Uses low-resolution texture features with a nonlinear kernel for classification



Evaluation Report:

Training Accuracy: 0.867

Testing Accuracy: 0.727

Training F1 Score: 0.867

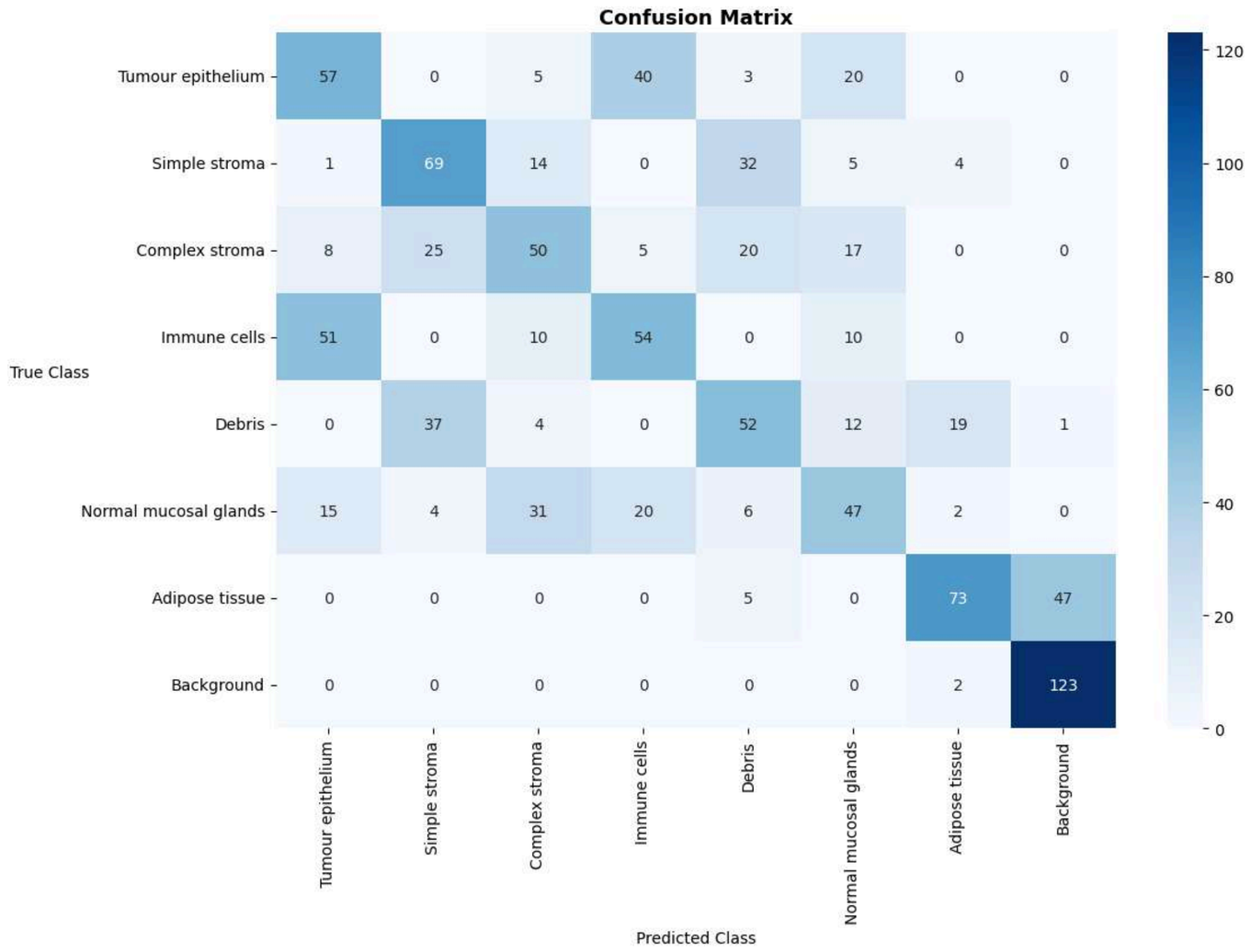
Testing F1 Score: 0.727

Overfitting Check: 0.140

Mild overfitting, with relatively balanced train–test performance

2 Downsample + Softmax

Model: Shows strong training performance with simple linear separation



Evaluation Report:

Training Accuracy: 0.9988

Testing Accuracy: 0.525

Training F1 Score: 0.9987

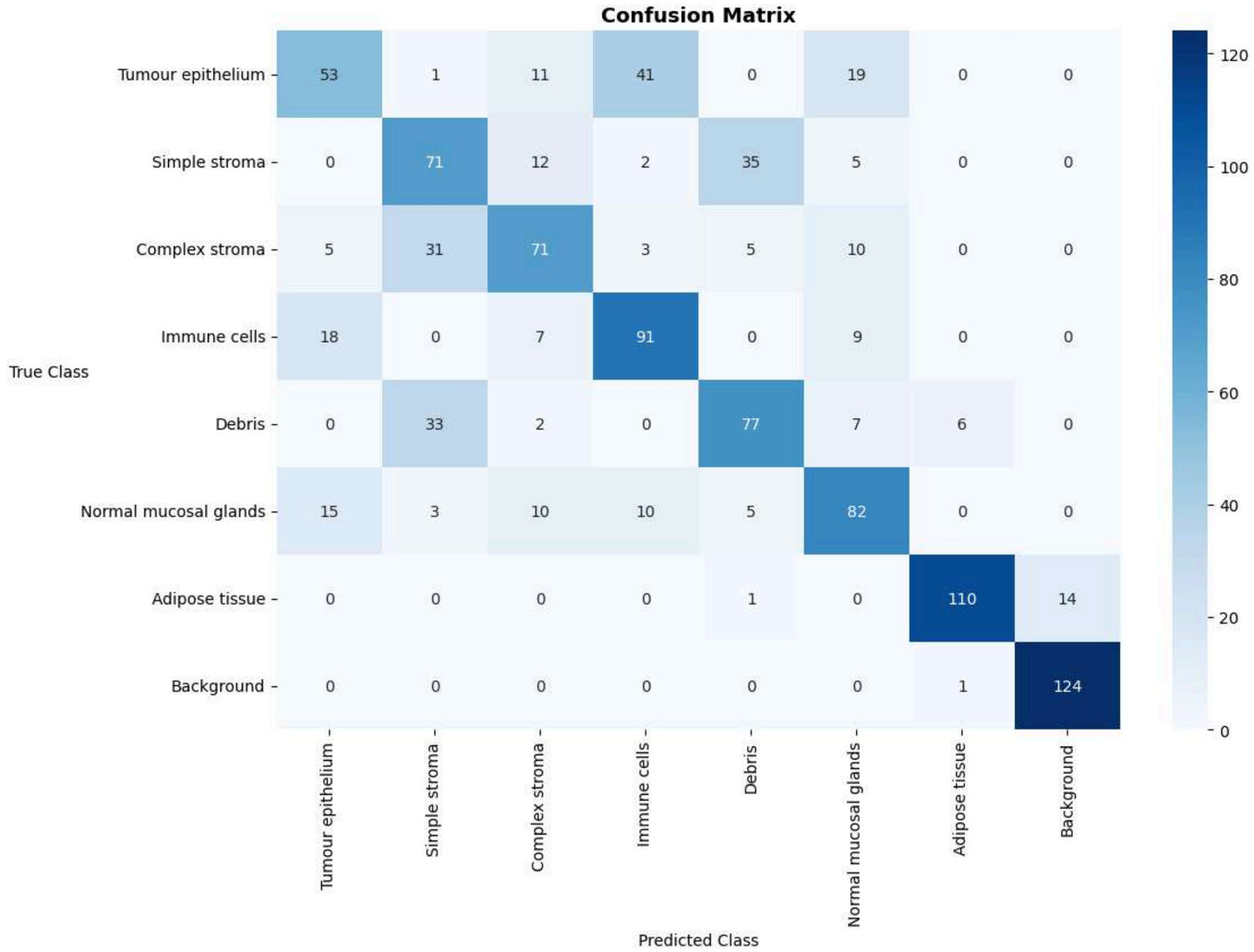
Testing F1 Score: 0.5177

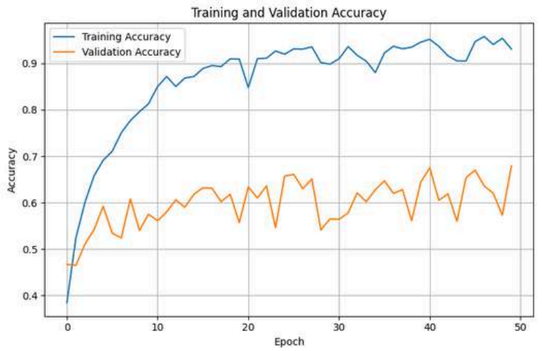
Overfitting Check: 0.479

Severe overfitting, weak performance on test data.

3 Downsample + NN

Model: Captures nonlinear relations and improves generalization





Evaluation Report:

Training Accuracy: 0.98

Testing Accuracy: 0.679

Training F1 Score: 0.9799

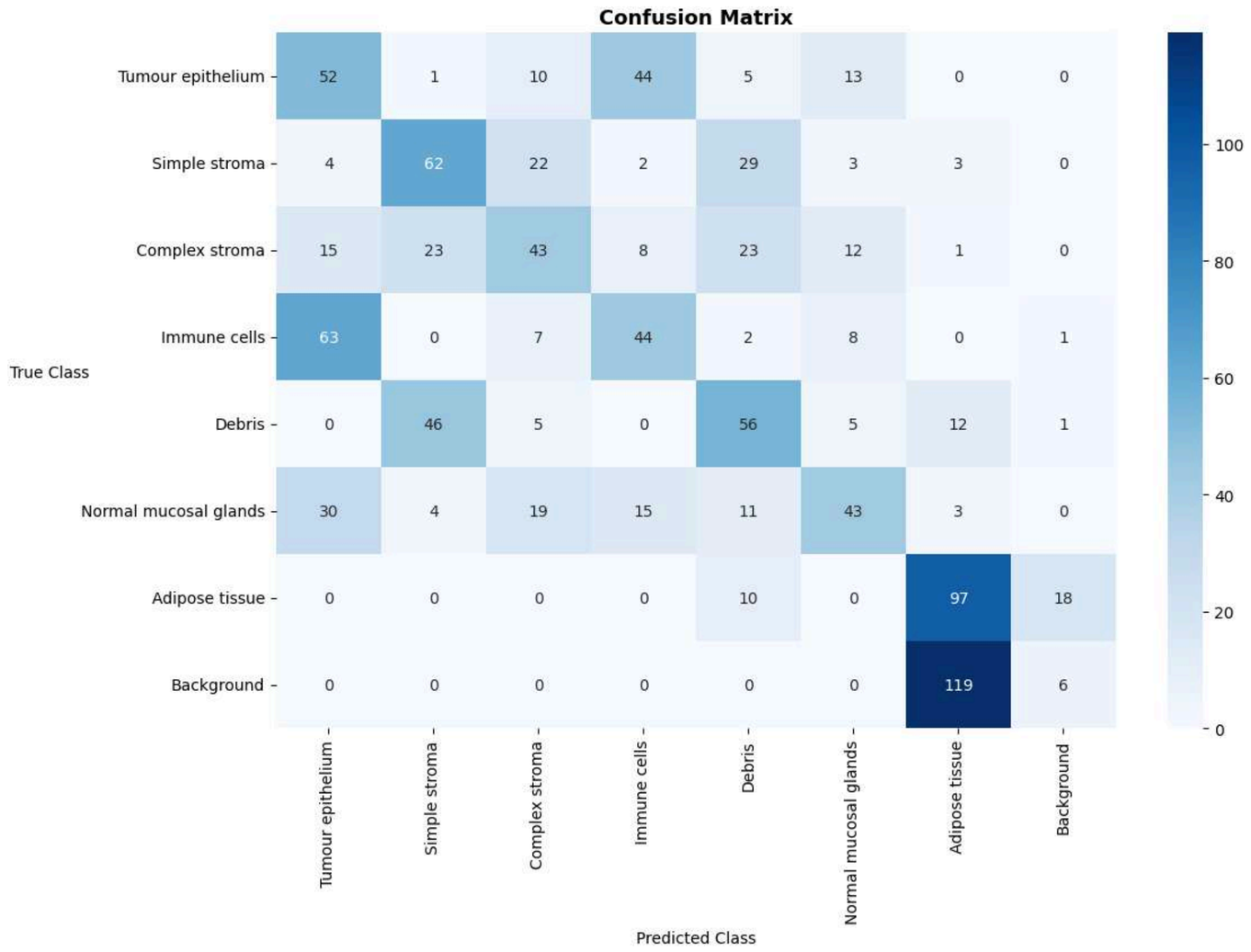
Testing F1 Score: 0.6769

Overfitting Check: 0.315

Moderate overfitting, with smaller train–test gap than previous models.

4 PCA + SVM (linear)

Model: Uses reduced features for linear separation



Evaluation Report:

Training Accuracy: 0.843

Testing Accuracy: 0.403

Training F1 Score: 0.8845

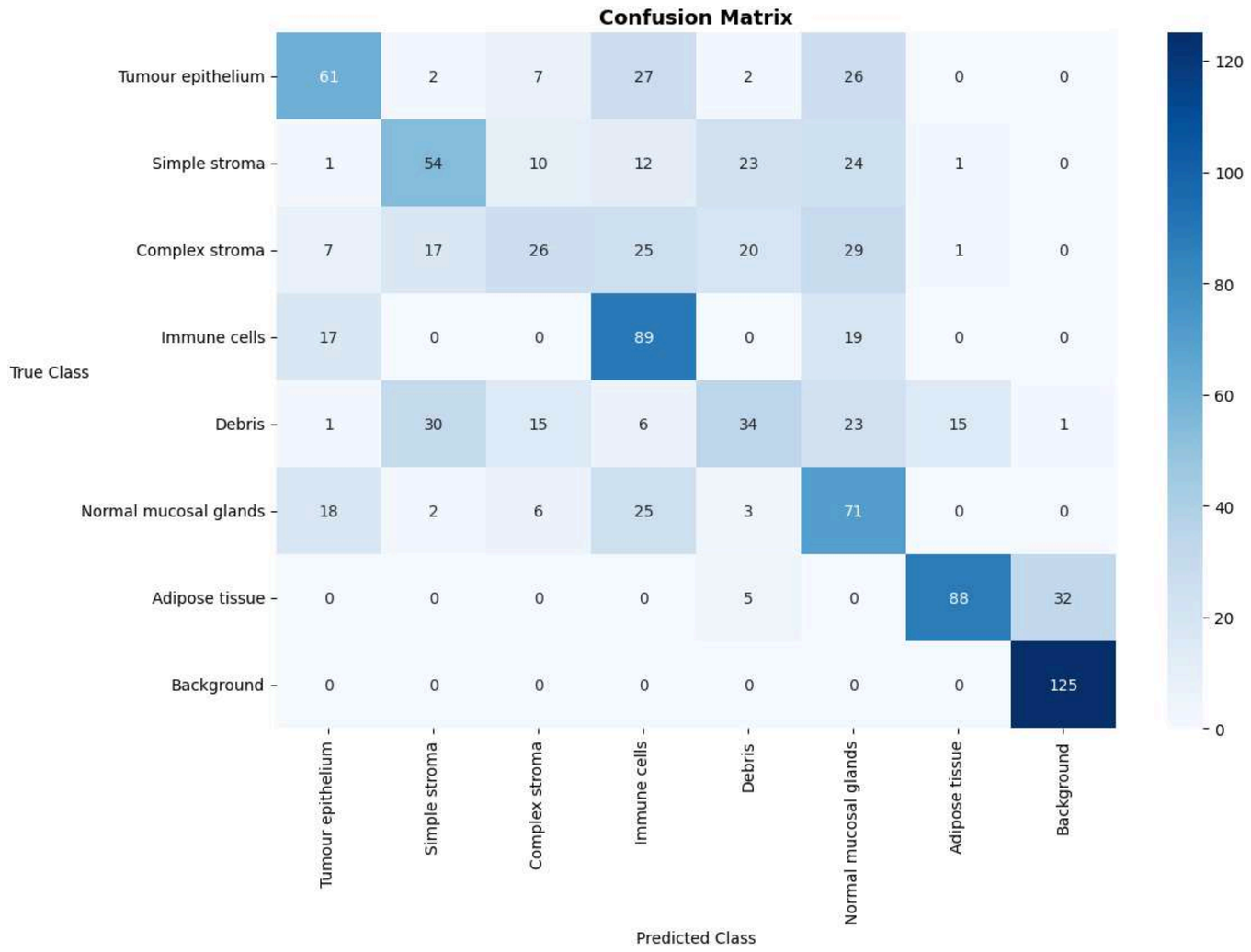
Testing F1 Score: 0.3795

Overfitting Check: 0.508

Mild overfitting, but overall weak generalization performance

4 PCA + SVM (RBF)

Model: Captures nonlinear structures in reduced space



Evaluation Report:

Training Accuracy: 0.9518

Testing Accuracy: 0.548

Training F1 Score: 0.9516

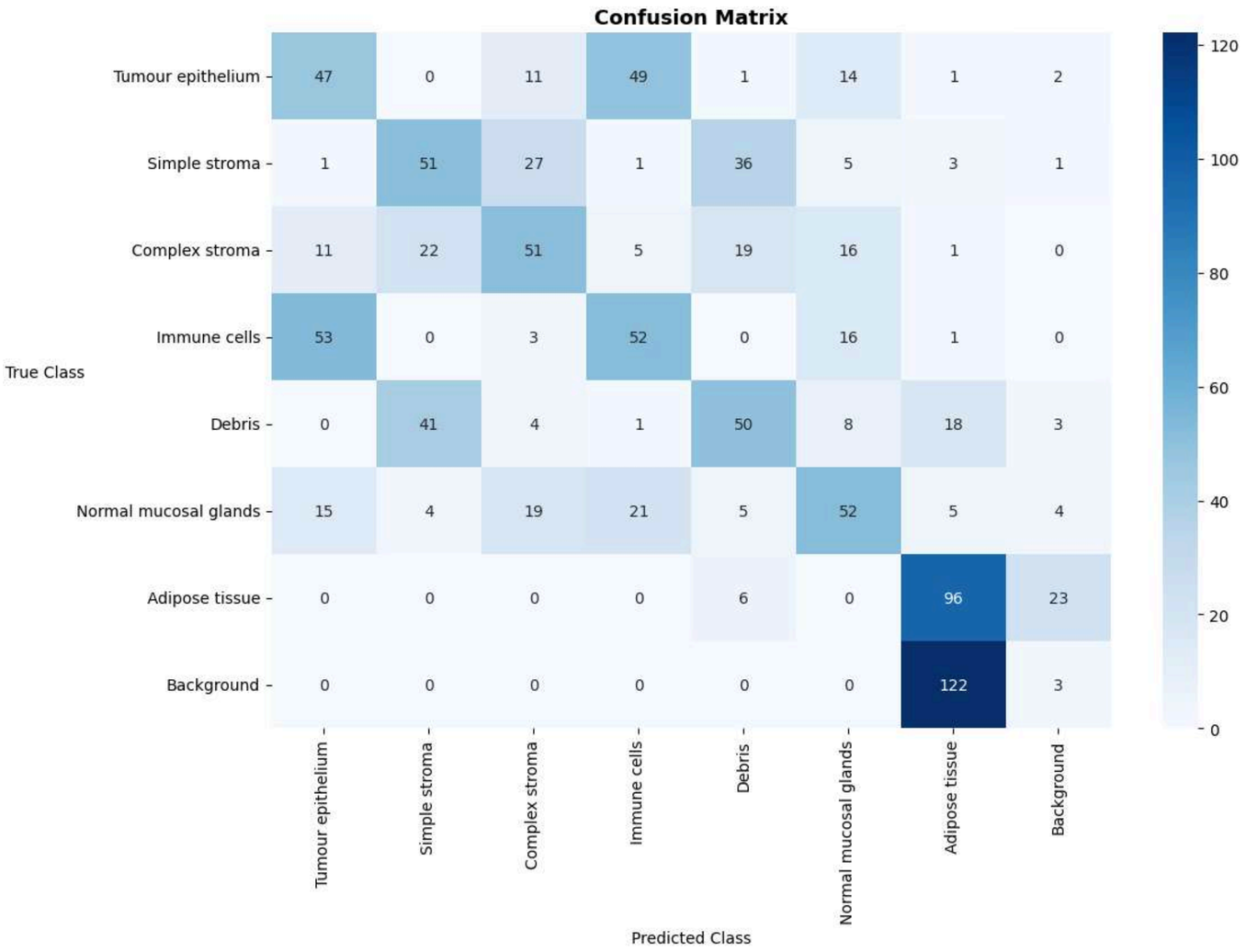
Testing F1 Score: 0.5336

Overfitting Check: 0.409

Noticeable overfitting, though better balance than the linear model

5 PCA + Softmax

Model: Applies linear classification on compressed features



Evaluation Report:

Training Accuracy: 0.7868

Testing Accuracy: 0.402

Training F1 Score: 0.7864

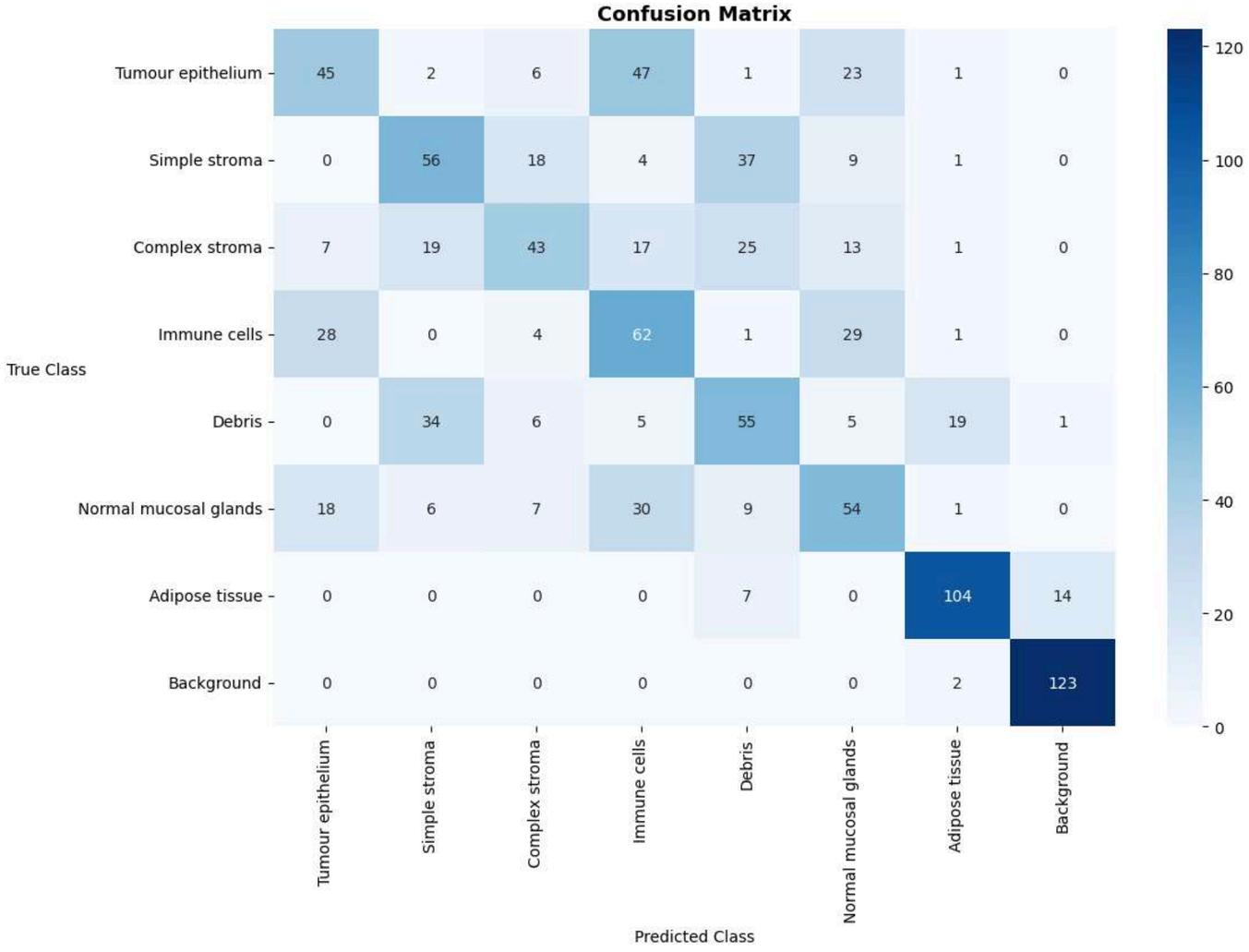
Testing F1 Score: 0.3793

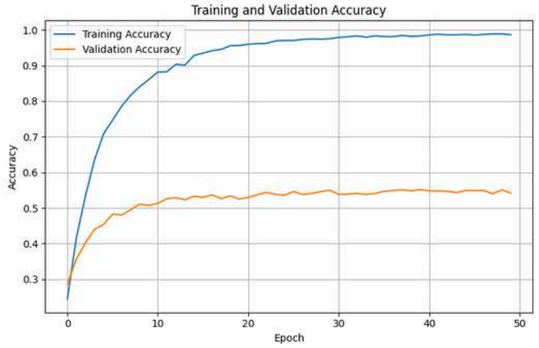
Overfitting Check: 0.379

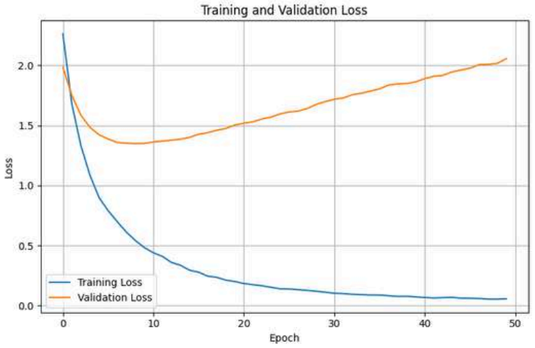
Underfitting dominates, with both train and test scores low

6 PCA + NN

Model: Leverages compressed nonlinear features for classification







Evaluation Report:

Training Accuracy: 0.98

Testing Accuracy: 0.679

Training F1 Score: 0.9799

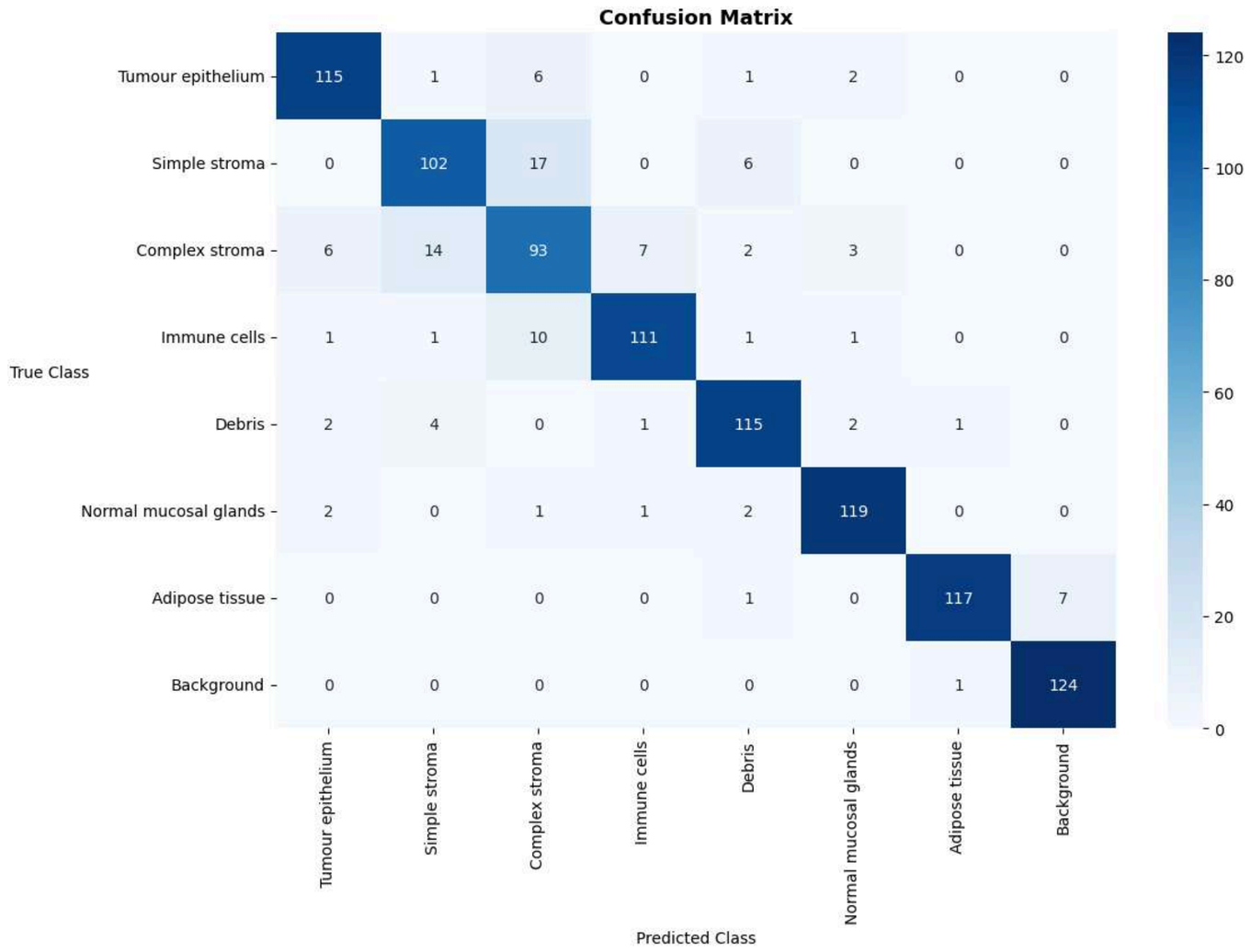
Testing F1 Score: 0.6769

Overfitting Check: 0.464

Moderate overfitting, yet stable and consistent performance

7 VGG16 + SVM (linear)

Model: Uses deep features for linear separation

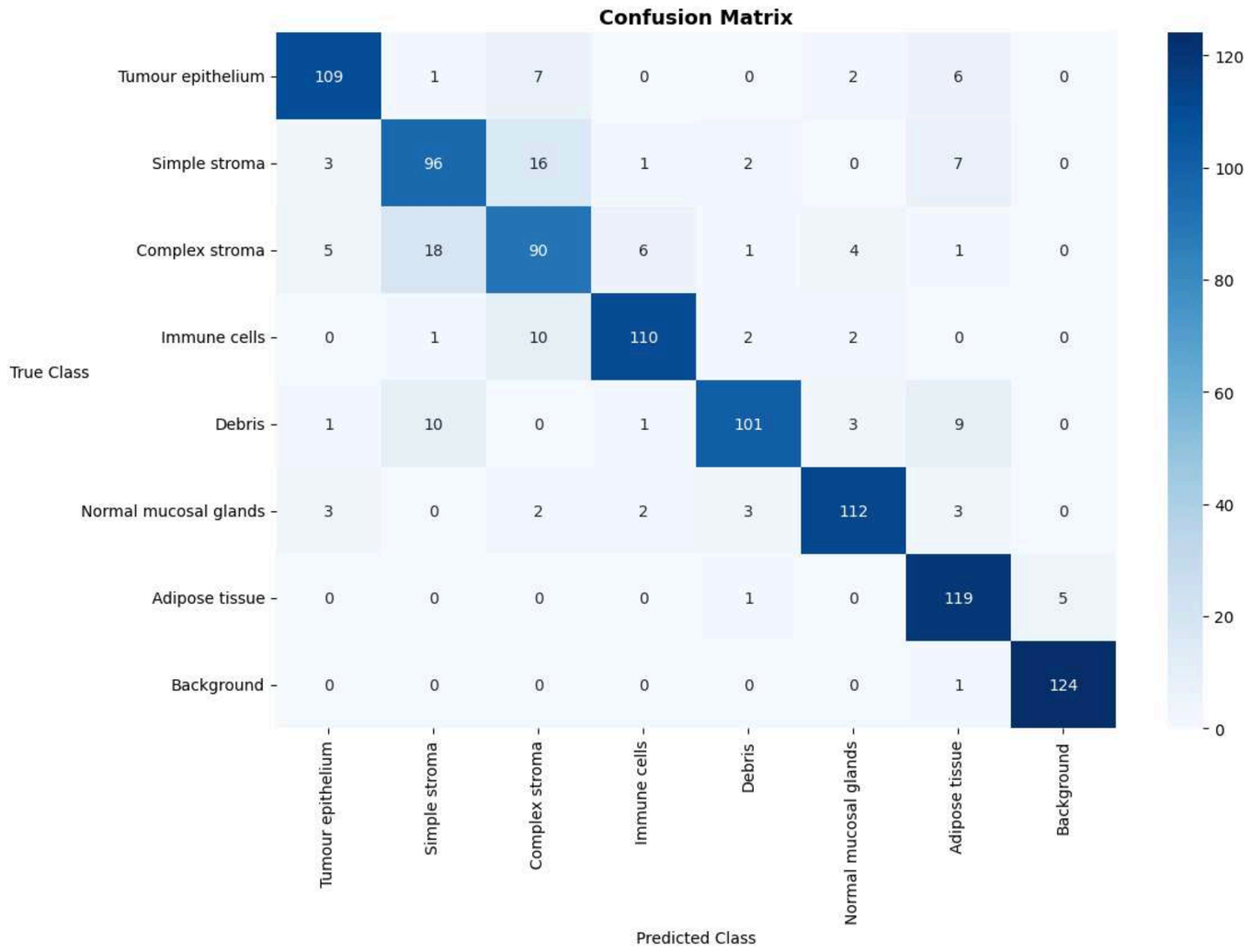


Evaluation Report:
Training Accuracy: 1.0
Testing Accuracy: 0.896
Training F1 Score: 1.0
Testing F1 Score: 0.896

Overfitting Check: 0.104
Minimal overfitting, with strong generalization across classes

7 VGG16 + SVM (RBF)

Model: Exploits deep nonlinear features for robust separation



Evaluation Report:

Training Accuracy: 0.9805

Testing Accuracy: 0.861

Training F1 Score: 0.9806

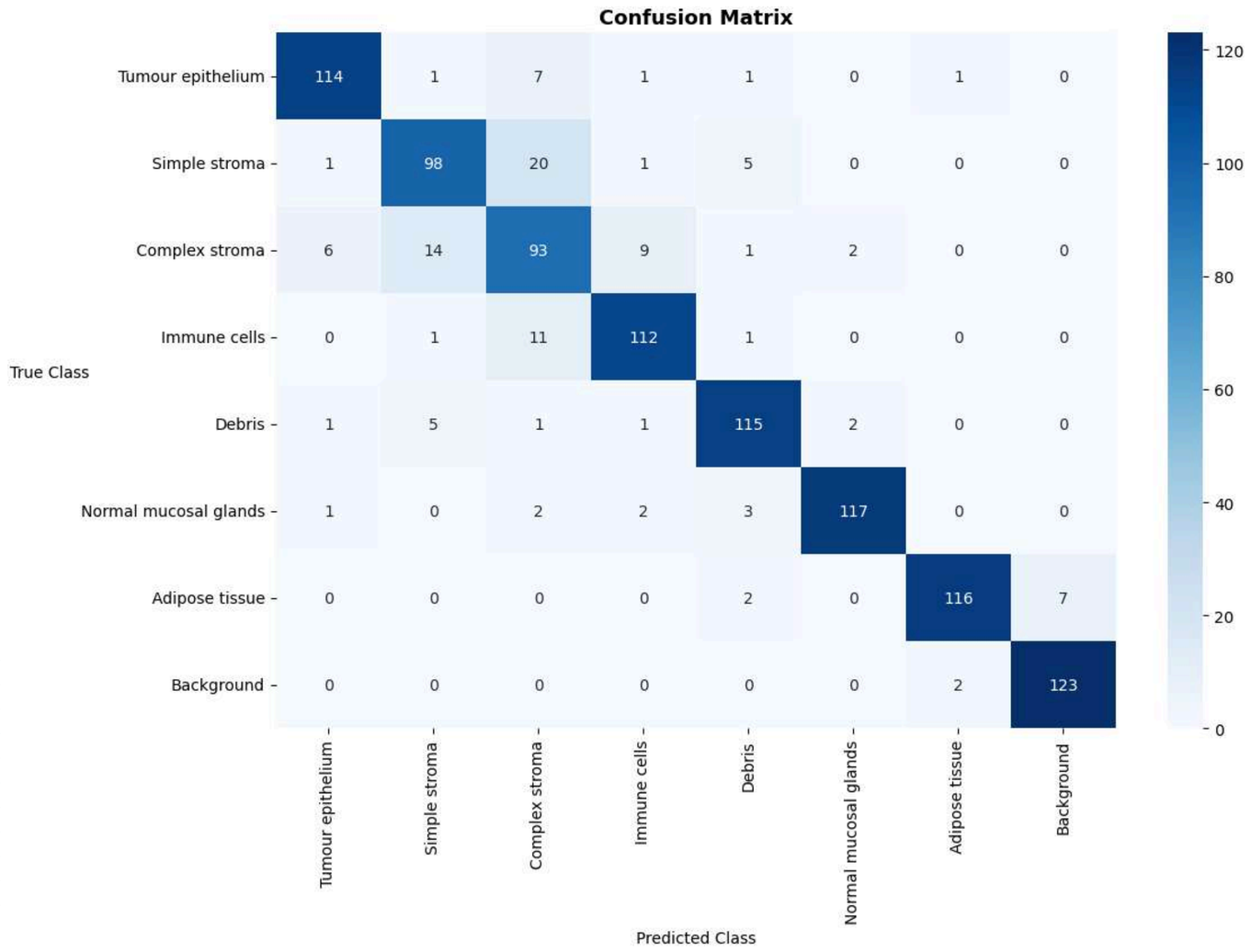
Testing F1 Score: 0.8608

Overfitting Check: 0.120

Slight overfitting, but maintains strong and balanced performance

8 VGG16 + Softmax

Model: Leverages deep learned features for strong classification

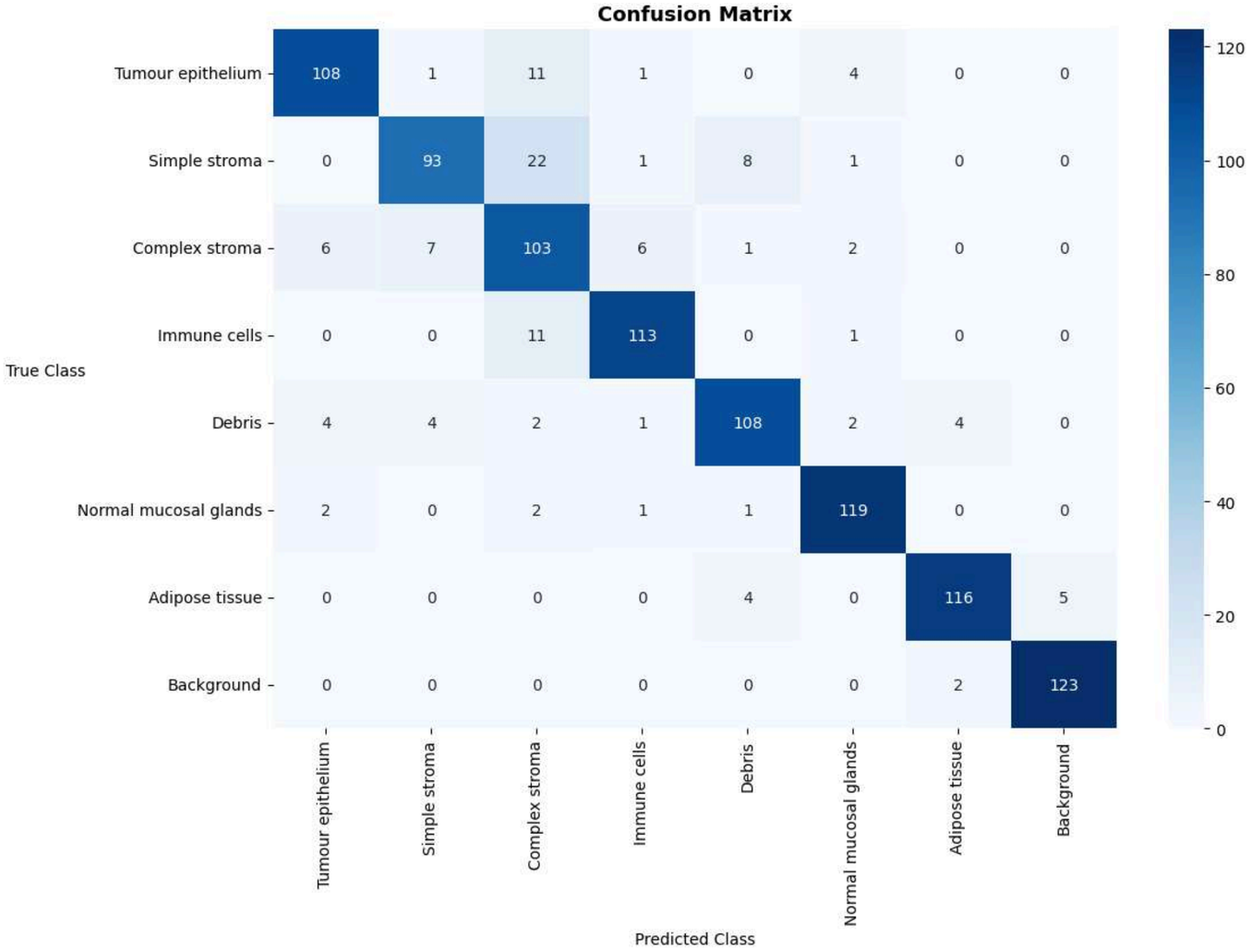


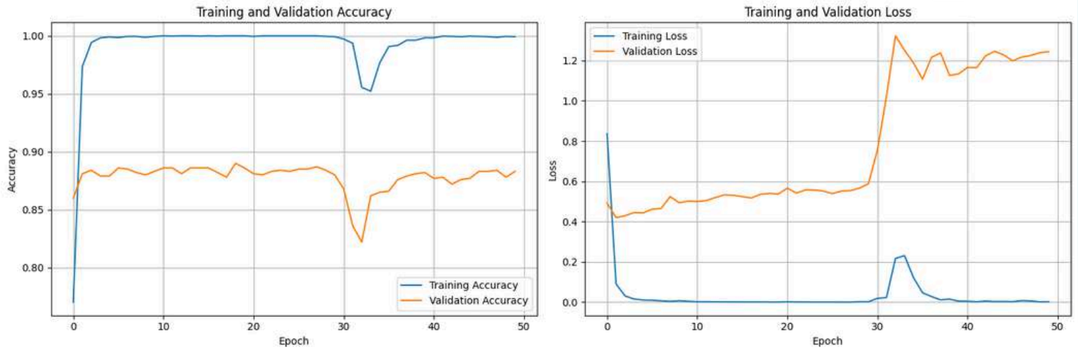
Evaluation Report:
Training Accuracy: 1.0
Testing Accuracy: 0.888
Training F1 Score: 1.0
Testing F1 Score: 0.8886

Overfitting Check: 0.112
Minimal overfitting, with excellent and stable test performance

9 VGG16 + NN

Model: Combines deep features with dense layers for refined learning





Evaluation Report:

Training Accuracy: 1.0

Testing Accuracy: 0.883

Training F1 Score: 1.0

Testing F1 Score: 0.8839

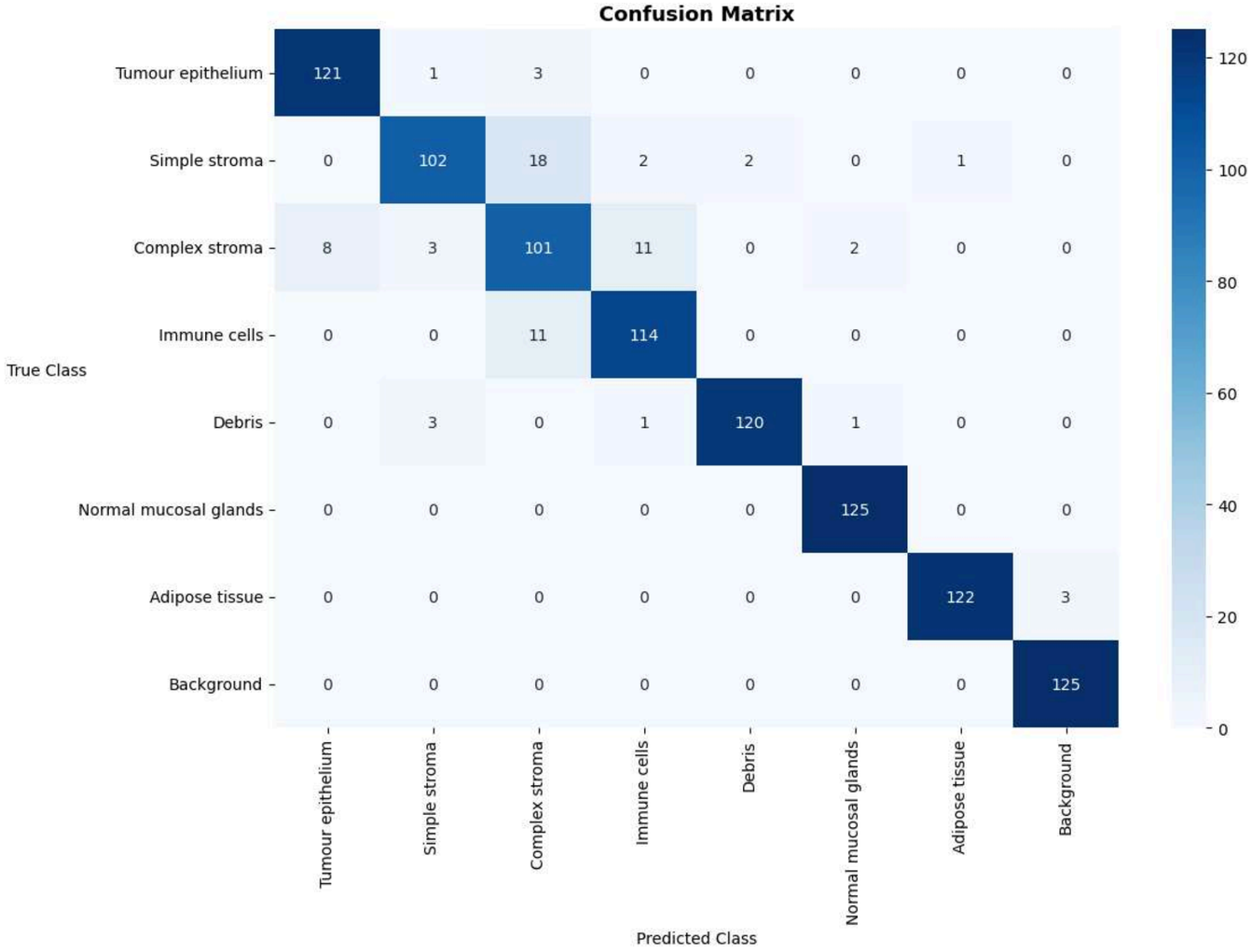
Overfitting Check: 0.123

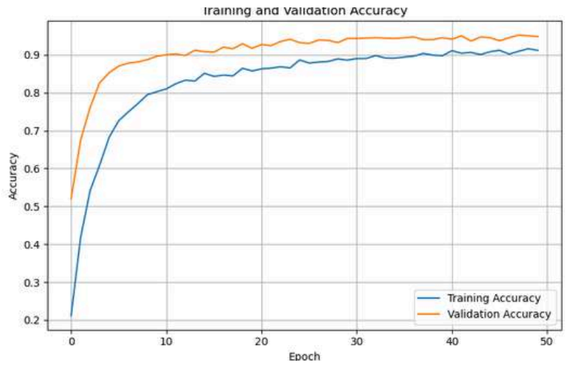
Slight overfitting, yet maintains strong and consistent test results

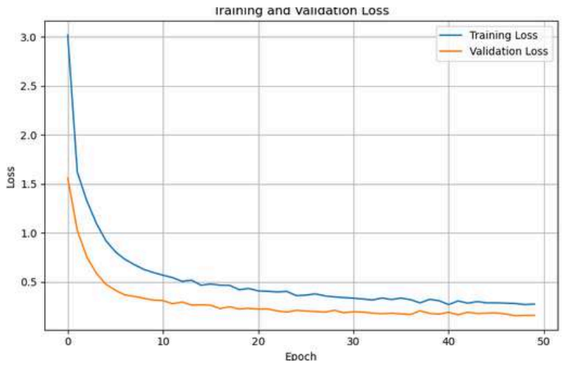
10 Fine-Tuned VGG16

1st Attempt

Model: Adapts deep layers for domain-specific optimization







Evaluation Report:

Training Accuracy: 0.927

Testing Accuracy: 0.930

Training F1 Score: 0.928

Testing F1 Score: 0.930

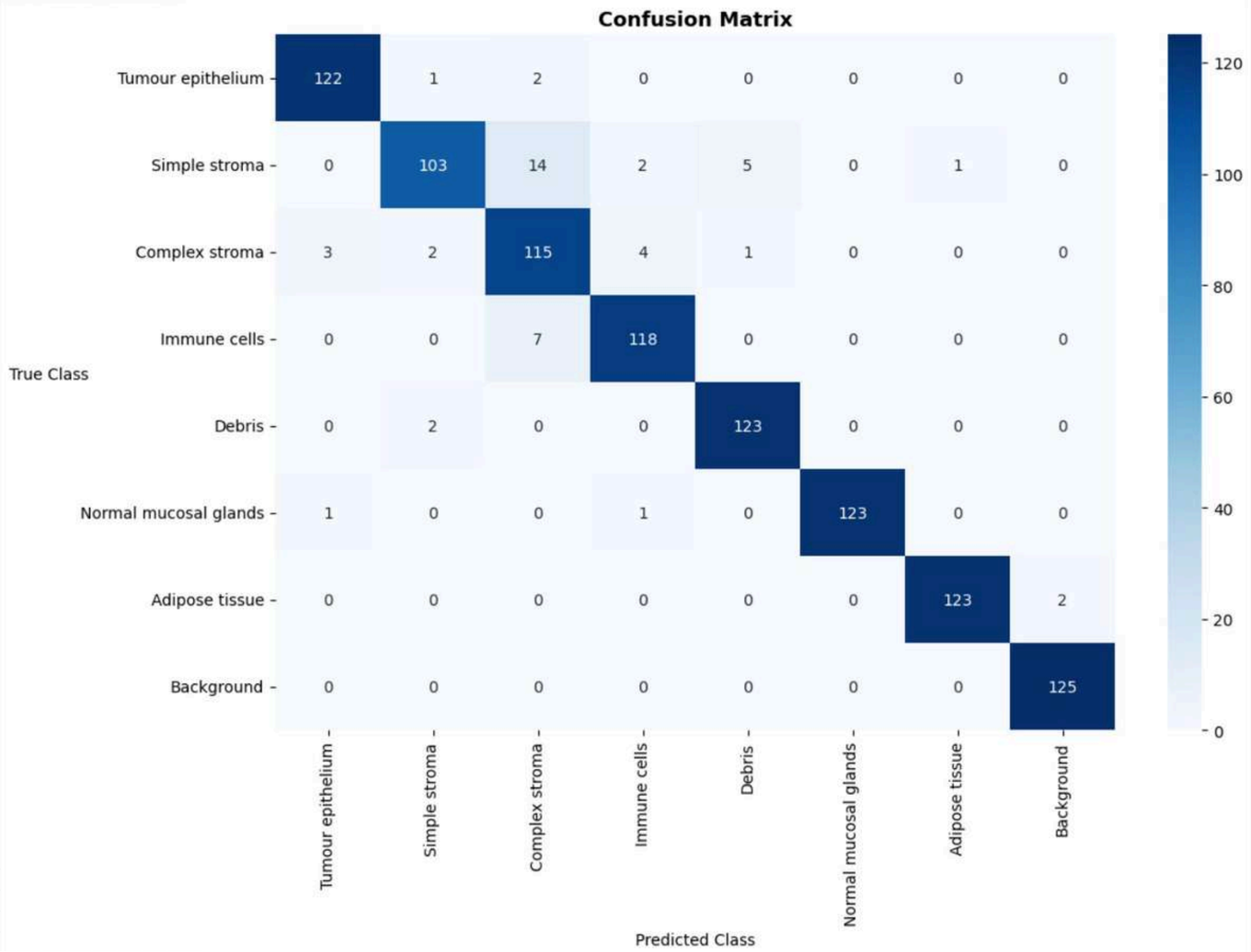
Overfitting Check: -0.003

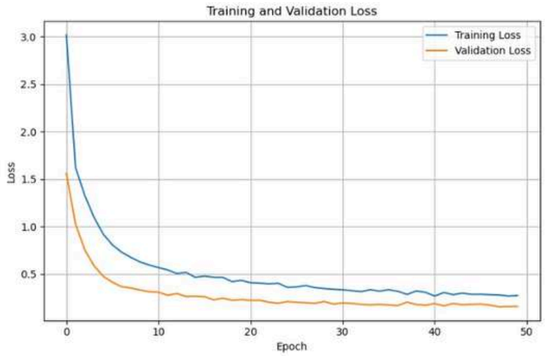
Good generalization -

Test performance matches training.

10 Fine-Tuned VGG16 2nd Attempt

Model: Adapts deep layers for domain-specific optimization





Evaluation Report:

Training Accuracy: 0.959

Testing Accuracy: 0.952

Training F1 Score: 0.959

Testing F1 Score: 0.952

Overfitting Check: 0.007

Almost Perfect Generalization - Test performance matches training.

10 Fine-Tuned VGG16

The following upgrades implemented to the final experiment.

Fine Tuning Methods:

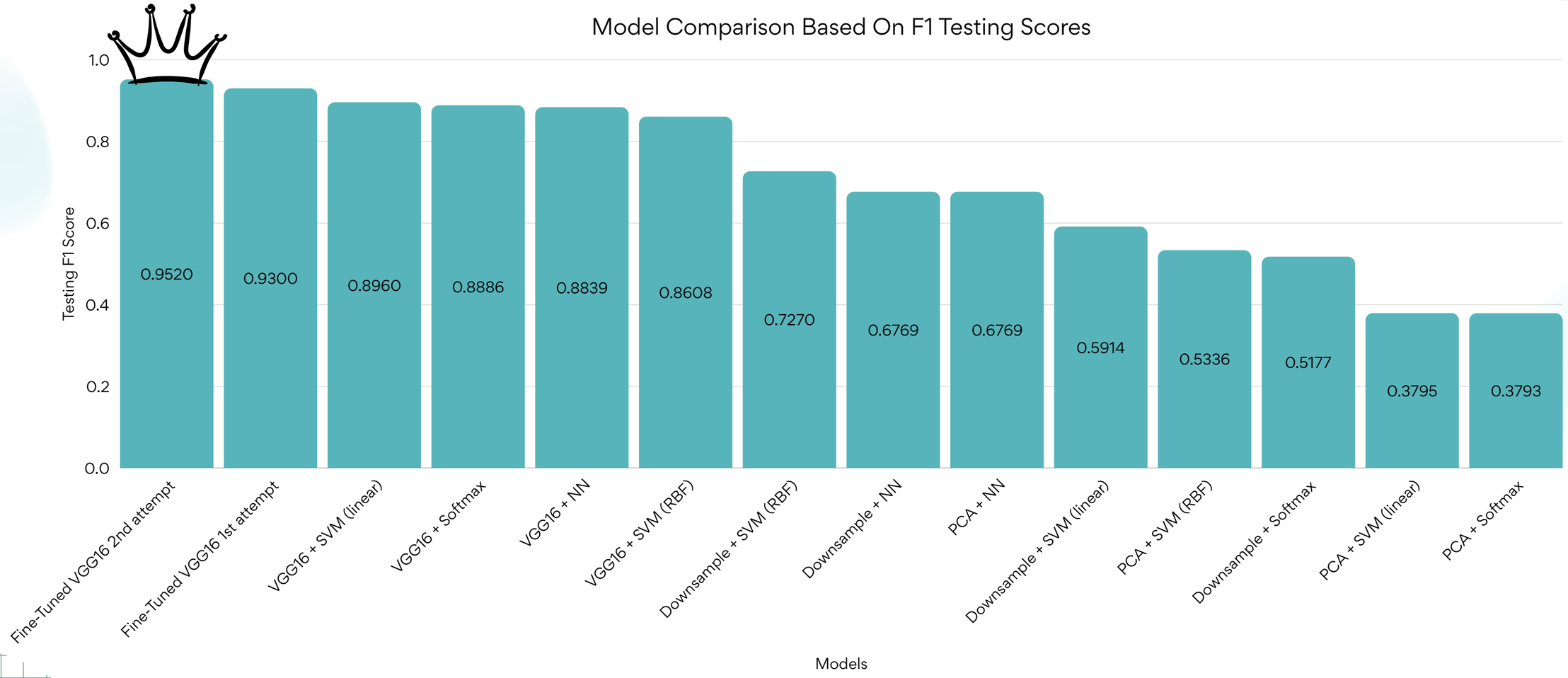
- Freezeing all the trained layers beside last four with adjusted weights.
- Setting the learning rate to 0.00001
- Using EarlyStopping method.
- Using ReduceLROnPlateau method..
- Using ModelCheckpoint method.

Data Augmentations Settings:

- Horizontal and vertical flips
- Rotation of ±45°
- Random scaling of image height and width up to 10% from the original size
- Filling empty areas after flipping
- Brightness adjustments
- Zoom in, zoom out
- Noise addition
- 150x150 Pixels (1st Attempt)
- 250x250 Pixels (2nd Attempt)

Model Review & Conclusions

Model Comparison



Technical Challenges

1st - Google Colab Re-Connectivity:

Using the Google Colab workspace for developing and training models led to continuous loss of connectivity and limited usage of workspace RAM and CPU, which prolonged the process.

Solution:

Switching to **Jupyter** workspace.

2nd - Uncertainty:

We didnt oversee the needed parameters for training the model and achieving a great results.

Solutions:

Understaning the dataset, Using Keras overflow, Test with different parameters.

Technical Challenges

3nd - Improving the final model:
Working with NN brings its own challenges since they are sensative to overfitting, specify with non-large datasets.

Solutions:
Using Data Augmentaion, Fine-Tuning the VGG16, Slow Learning Rate and Using Callbacks.

4th - Reserved duty:
We both were on reserve duty for a long period, missed several lessons, and fell behind schedule.

Solutions:
Coping strategies included the use of Ritalin, alcohol, and cigarettes (Eli), as well as downloading the lectures for offline review.

Future Recommendations

Future improvements aim to enhance generalization, robustness, and real-world applicability of the models

1

Improving Overfitting Control

Use stronger regularization, richer data augmentation, and broader cross-validation.

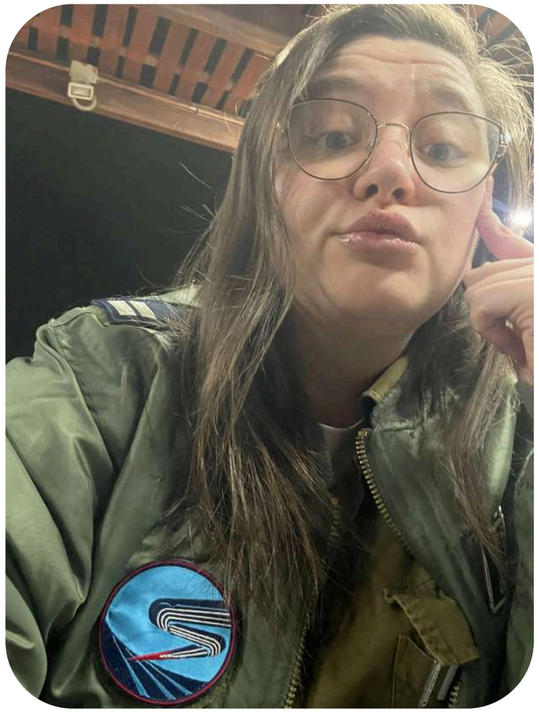
2

Expanding Data and Balance

Add more datasets, balance classes, and include relevant clinical features.

3

Using advanced models and transformer architectures.

Explore advanced deep learning approaches, including transformer-based architectures, to further improve model performance and generalization.

Thank You